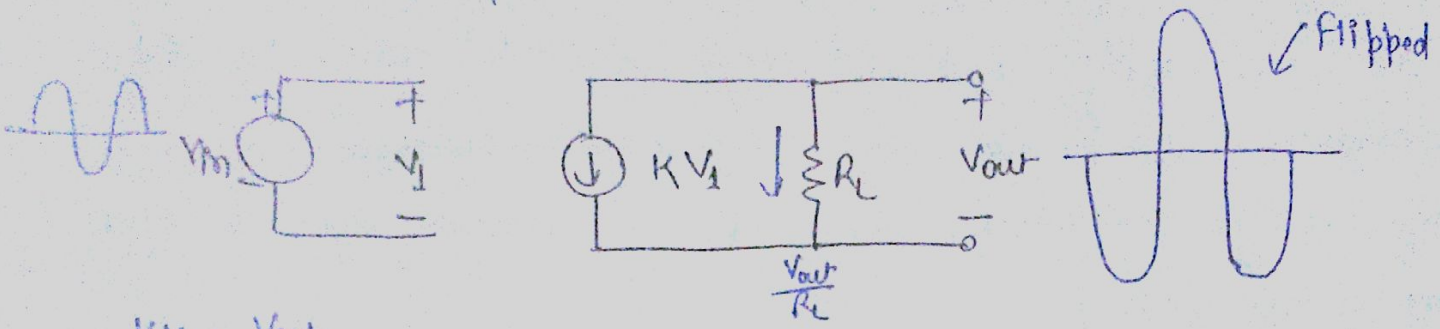
# Lecture - 29

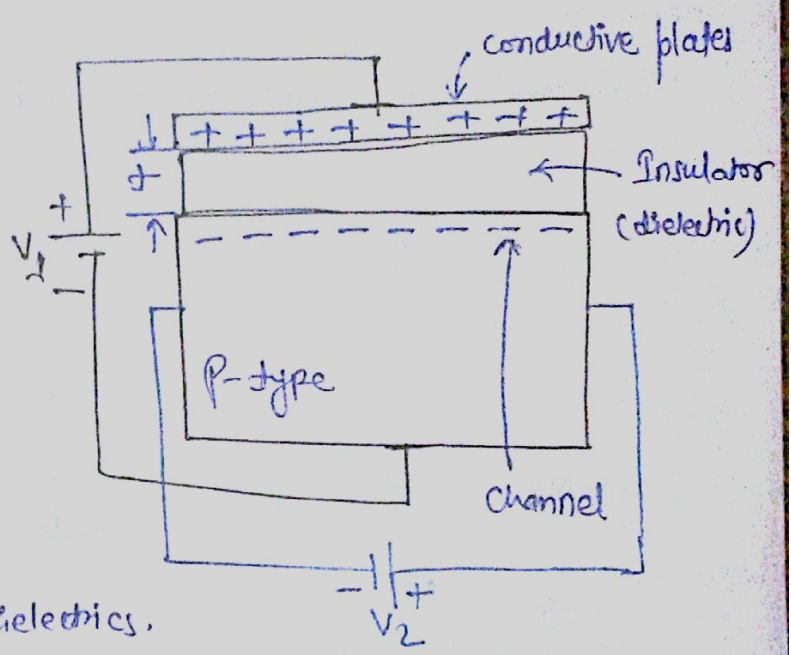
- Let's build an amplifier:-




$$KV_1 + \frac{V_{out}}{R_L} = 0$$

$$\frac{V_{out}}{R_L} = -KV_{in} \Rightarrow V_{out} = -KR_L V_{in}$$

## Observations:-




→ for building a capacitor, we take a plate which is conductive and another conductive plate and that forms a capacitor and between the plates we have something could be air or dielectrics.

* In this we use a conductive plates and p-type semiconductor since p-type semiconductor has some conductivity so it will act as conductive plates. In between them we place dielectric. We apply a voltage source across it. Positive charges are induced on conductive plates. Negative charges are induced on p-type semiconductor, the layer of negative charges are called as channel.

* $Q = CV_1$

If $V_1 \uparrow \Rightarrow Q \uparrow \Rightarrow$ Electron density increases.

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* If $t$ (thickness) $\downarrow$, $V_1$ (constant), $\Rightarrow C \uparrow \Rightarrow Q \uparrow$

$\Rightarrow$ Electron density in the channel also increases.

* If we apply another voltage source $V_2$ across the p-type semiconductor then we have some free electrons here and we have voltage difference across it so electrons begin to flow through the battery.

* If $V_1$ increases, electron density increases,

$\Rightarrow$ Resistance between channel decreases, current flow through channel increases.

Therefore we have voltage dependent current source.

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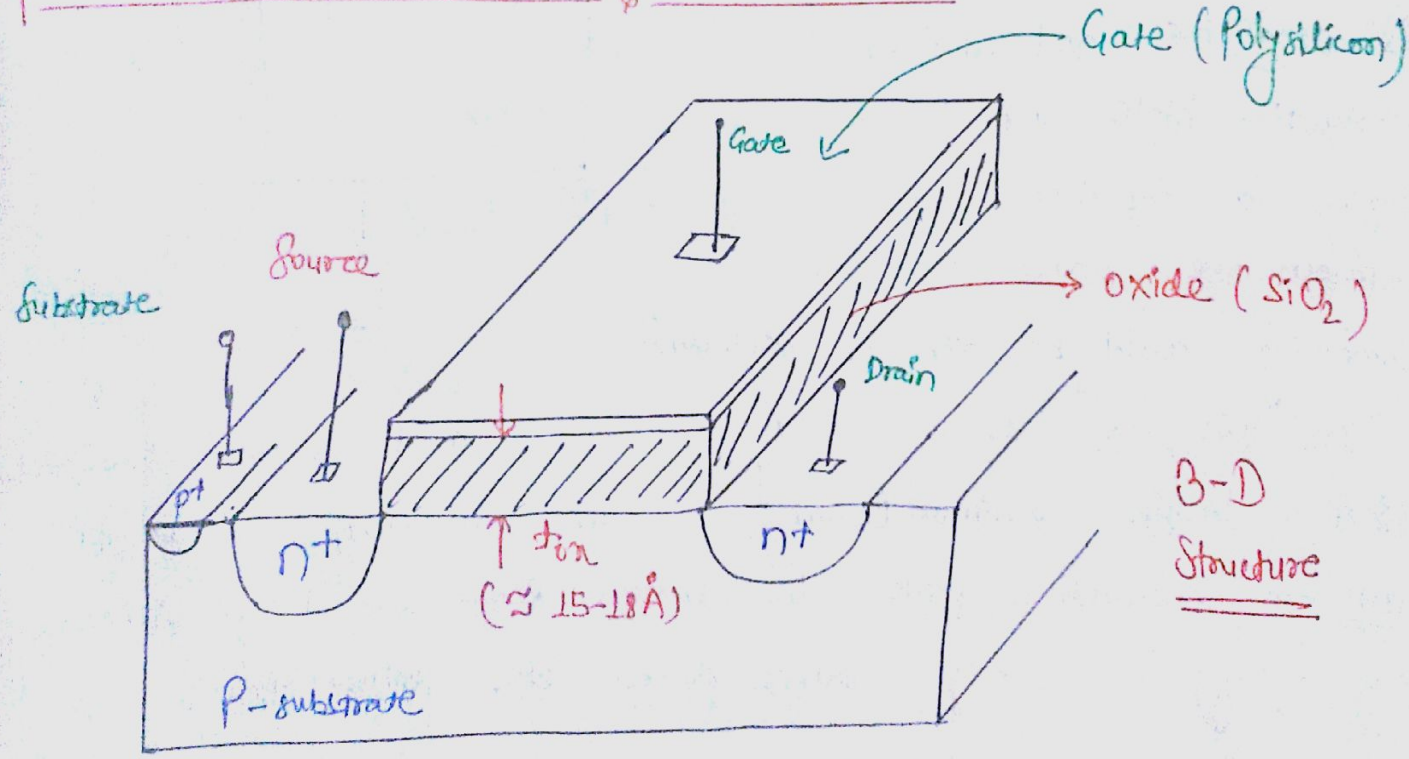
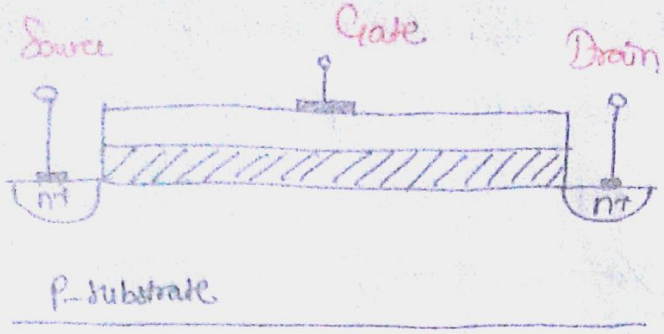
## Metal oxide semiconductor (MOS) structure:-

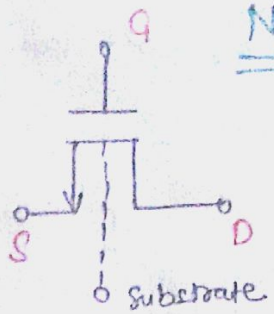



3-D Structure

(1) The MOSFET (Field effect transistor) has four terminals, source, gate, drain and substrate.

(2) The MOS structure is symmetric [BJT is not symmetric].

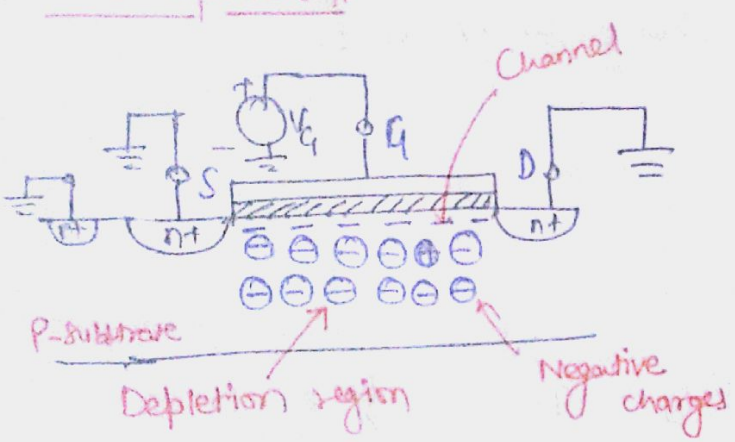


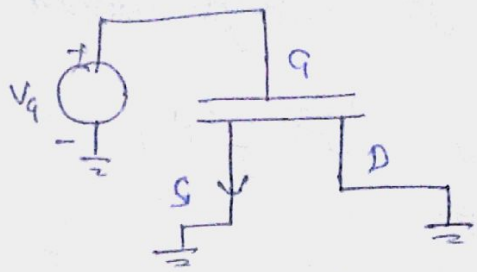




NMOS

* The current through source and drain is controlled by Gate voltage.

MOS operation:-




$V_G = 0.2V$

→ If we connect ground to source and drain, one voltage source $V_G$ to gate. When $V_G$ is slightly positive. Gate becomes more positive it will repel holes present in the p-substrate. Therefore at remaing space we have -ve charges. These negative ions become more and more in population as the voltage at gate increasing. We got a depletion layer here.

→ Once $V_G$ reaches the threshold voltage $V_{Th}$ then free electrons goes to form channel for conduction process. That means device stops exposing negative ions and instead bring free electrons.

→ Below the $V_{Th}$, we don't have electrons therefore doesn't have channel. Once the Gate voltage passes the threshold $V_{Th}$ we have some free electrons that causes to flow current through MOS.

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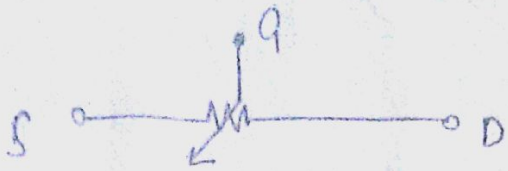
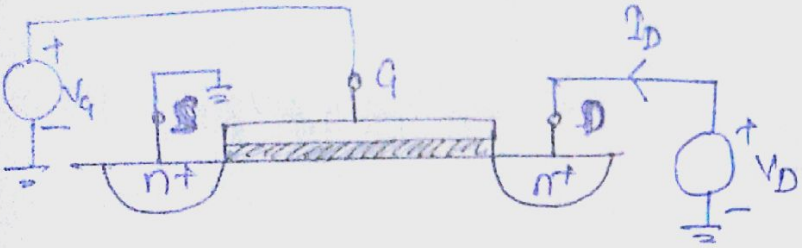
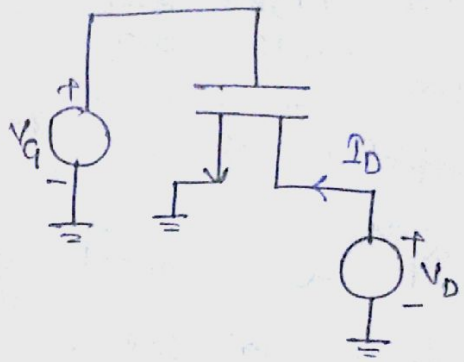
* As the $V_G$ increases beyond $V_{th}$, electron density in the channel increases ⇒ resistance between S and D ↓




Source and drain resistance is variable resistance which have another terminal gate, which can alter the value of resistance. If the gate voltage is low, resistance becomes high and if gate voltage is high, resistance have lower value.

* Assume $V_D > V_S = 0$,

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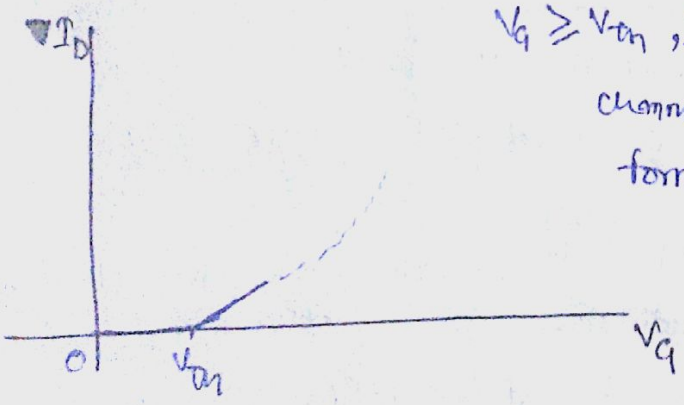
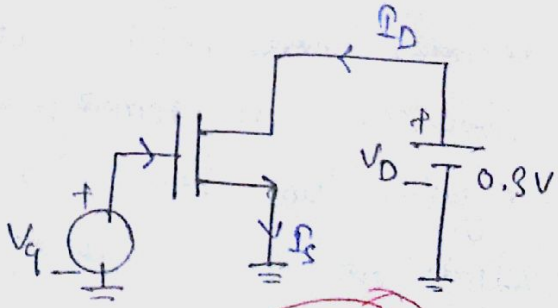
Case I $V_D$ = constant, eg, 0.3V




$V_G \geq V_{th}$, $I_D$ flows

channel forms.



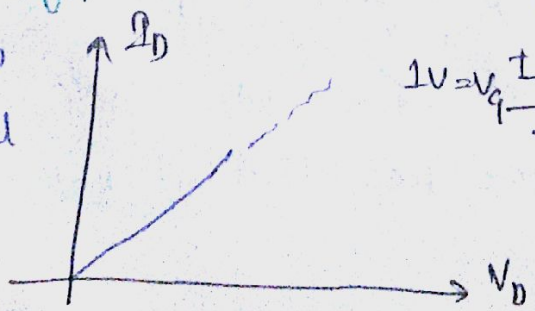

$I_S = I_D$

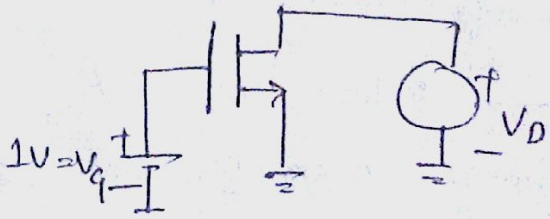
$I_G = 0$

Case II $V_G$ = constant, e.g., 1V.

when $V_D = 0$, $I_D = 0$

since $V_G > V_{th}$ so channel formed, so source & drain act like resistance.

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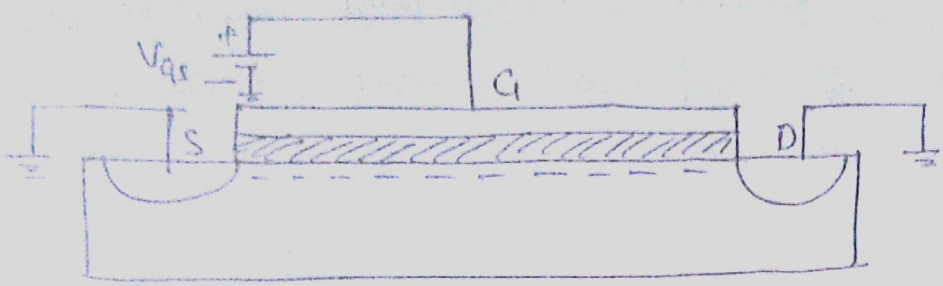
# Lecture - 30

## Behaviour of channel :-

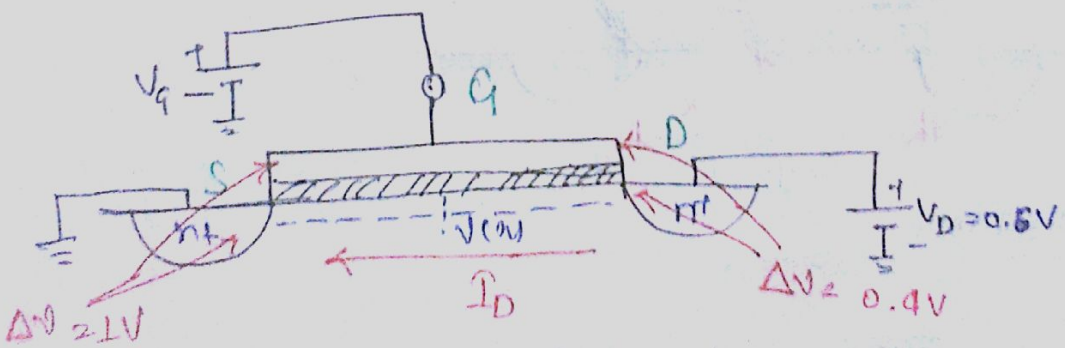
**Case I** $V_{GS} > V_{th}$, $V_S = V_D = 0$




→ Turned ON, but does not have current. Because Drain and source voltage is same i.e. zero.

→ If we find the density of electrons at different points of channel, we get equal density at all points because we have capacitor and charge accumulates on the plates of capacitor is in proportion to the voltage that we have. We have same voltage difference at different points of channel so same charge density we ~~[illegible]~~ got.

**Case II :** $V_{GS} > V_{th}$, $V_S = 0$, $V_D > 0$




→ Since we have channel to flow current. The channel acts like a resistance. Therefore current flowing through the resistive path (many resistances). Because we have 0.5V at Drain and 0V at the source, and we have a resistive path the voltage varies from 0.5 to 0 with respect to 0 if we

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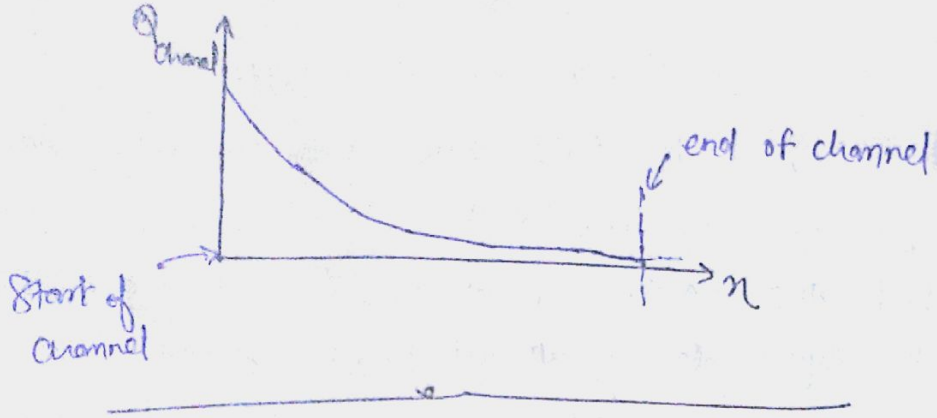
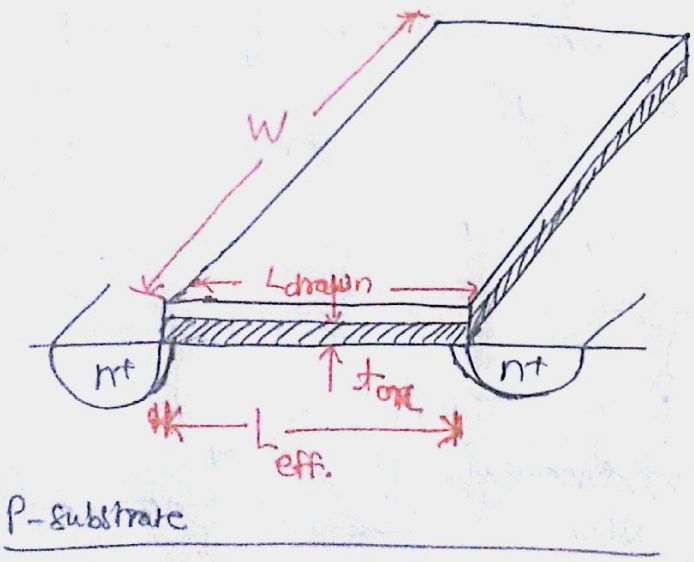
trend from drain to source.

→ Voltage along the channel increases as we go from source to drain.

→ The electron density of the channel varies from source to drain because voltage across the local capacitor varies from source to drain (decreases).




• Dimension of MOSFET :-




→ When we draw source and drain areas this doesn't start from the edge. During fabrication, when we implant source and drain region in ~~semiconduction~~ semiconductor process then heats this up to source and drain region penetrates and diffuses and little bit overlap Gate Area.

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→ $L_{drawn}$ is the drawn length when we draw gate region and $L_{eff}$ is length when the device is fabricated which differs by some amount from $L_{drawn}$.

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## I-V characteristics:-

### ① Channel charge density:-

**Case I:** $V_{GS} > V_{Th}$, $V_D = 0$

$$Q_{ch_{total}} = \underbrace{WL}_{\text{Capacitance}} \cdot \underbrace{C_{ox}}_{\text{Capacitance per unit Area (farad/m}^2)} \overbrace{(V_{GS} - V_{Th})}^{\text{overdrive voltage}} \quad \text{Coulomb}$$

→ If $V_{GS} = 0$, $Q_{ch} = 0$ which is true. But if $V_G = 0.1V$ then its is less than threshold volt $V_{Th}$ so we don't have channel so no $Q_{ch}$ should be there but because of this we get some charge so its not true for correcting this we just subtract $V_{GS}$ by $V_{Th}$.

$$\boxed{Q_{ch_{density}} = WC_{ox}(V_{GS} - V_{Th})} \quad \text{Coulomb/m}$$

**Case II:** $V_{GS} > V_{Th}$, $V_D > 0$

$$\boxed{Q_{ch_{density}} = WC_{ox}\left[V_{GS} - V_{Th} - \overbrace{V(x)}^{\text{voltage function of } x}\right]} \quad \text{coulomb/m}$$

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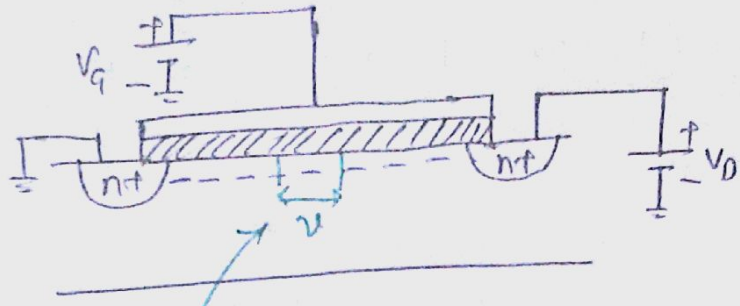
## ② Drain Current :-

→ In a general semiconductor, with some doping, when we apply an electric field we obtain a current and we know that when we apply an electric field that the carriers reaches certain velocity.

$$v = \mu E$$

$$\because \quad E = -\frac{dv}{dx}$$

$$\Rightarrow \quad v = -\mu \frac{dv}{dx}$$




Total charge in this length

$$I_D = -v \cdot Q_{ch\,density} = -v\, W C_{ox} [V_{GS} - V_{Th} - V(x)]$$

in one second

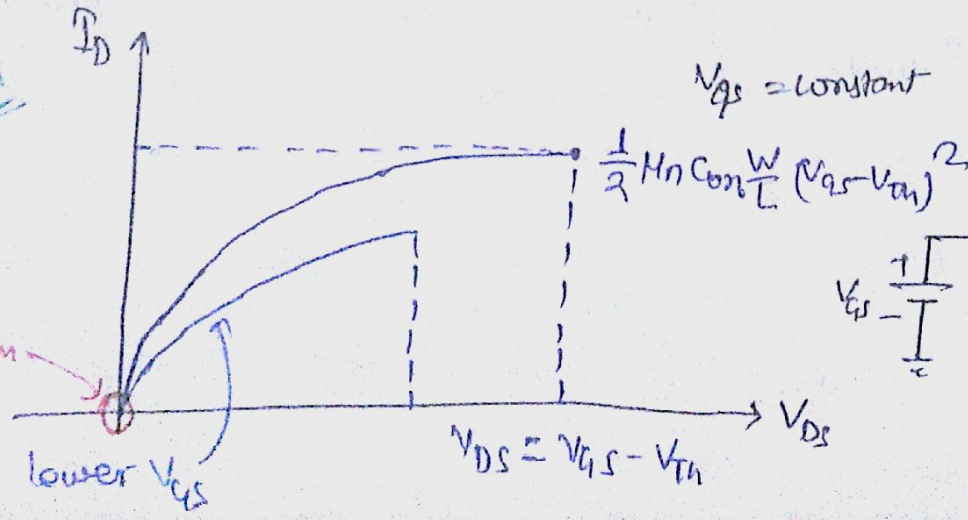
$$I_D = +\mu_n \frac{dv}{dx} C_{ox} W [V_{GS} - V_{Th} - V(x)]$$

$$\Rightarrow \int_0^{L_{eff}} I_D \cdot dx = +\mu_n C_{ox} W \int_0^{V_{DS}} [(V_{GS} - V_{Th}) - V(x)]\, dV$$

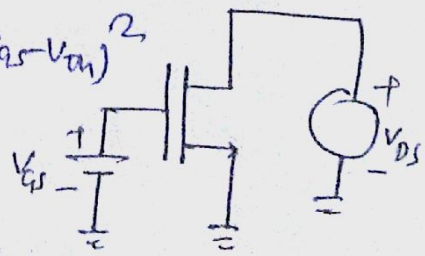
$$\Rightarrow I_D \cdot L_{eff} = +\mu_n C_{ox} W \left[ (V_{GS} - V_{Th}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$\Rightarrow \boxed{I_D = +\frac{\mu_n C_{ox} W}{2 L_{eff}} \left[ 2 V_{DS} (V_{GS} - V_{Th}) - V_{DS}^2 \right]}$$

Valid only if $V_{GS} \geq V_{Th}$




$I_D = 0$
but $V_{GS} > V_{Th}$
$V_{DS} = 0$
Device ON

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→ The $I_D$ current reaches its max. value when $V_{DS} = V_{GS} - V_{th}$.

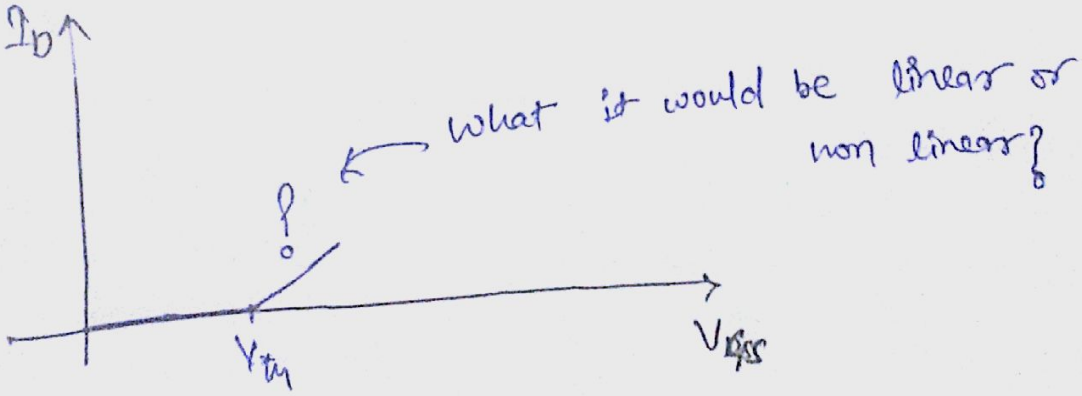
→ If $V_{GS}$ is lower then the $I_D$ has max value does not have equal to previous it will be lower. If $V_{GS}$ is higher then we get higher peak than previous.

→ The current $I_D$ is proportional to $\mu_n$ mobility of electron because if mobility increases then velocity increases so current increases.

→ The larger the $C_{ox}$ then larger the $I_D$ because of increase in $C_{ox}$ increases charges so current also increases.

→ The width increases means becomes wider therefore can have larger current so it is proportional to W.

→ If device is longer then the resistance becomes longer so it has higher resistance so current reduces that's why the $I_D$ is inversely proportional to L.

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